ANISH S. PATIL

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PROFESSIONAL SUMMARY

Software Engineer transitioning into Data Science and AI/ML, with hands-on experience building Retrieval-Augmented Generation (RAG) and AI infrastructures for practical, real-world use cases. Proficient in Python, machine learning, data preprocessing, LLMs, embeddings, vector databases, and LangChain, with familiarity in developing AI-driven workflows for testing and production use cases. Brings additional software quality experience through defect tracking, client deliverables, and collaboration with QA teams and project managers.

TECHNICAL SKILLS

- **Programming**: Python, SQL
- **Libraries**: NumPy, Pandas, Matplotlib, Scikit-learn
- **Machine Learning**: Supervised Learning, Unsupervised Learning, Machine Learning Applications, Reinforcement Learning, Preference Optimisation, Model Evaluation
- **Gen AI :**: LLMs, RAG Systems, Prompt Engineering, Embeddings, Vector Databases, LangChain, Generative AI and LLM Applications, Agentic AI, LangGraph
- **Tools**: Jupyter Notebook, VS Code, Claude Code, Code X, Github Copilot, Docker
- **Version Control**: Git, GitHub
- **Databases**: Oracle SQL, MySQL

PROJECTS

RAG-Based AI Video Assistant| Python, NLP, RAG, Whisper, LLMs

- Developed a generative-AI video assistant using Python and Retrieval-Augmented Generation (RAG) to process and analyze video content.
- Implemented speech-to-text transcription using Whisper to convert video audio into text.
- Implemented Hindi-to-English translation, timestamp-based video retrieval, and context-aware LLM search for accurate video segment extraction.
- Enabled context-aware search using LLMs for accurate video segment extraction.
- Optimized video processing workflow using FFmpeg for efficient audio extraction and preprocessing.

Object Detection using Drone | Python, TensorFlow, OpenCV, COCO Dataset

- Developed and evaluated a real-time object detection application using TensorFlow, OpenCV DNN, and SSD MobileNet V3 for live video streams.
- Utilized pretrained COCO dataset models to detect 80+ object classes including person, vehicles, and daily objects.
- Integrated computer vision with hardware (GPIO-based servo motor) to simulate object tracking for drone applications.
- Processed live video feeds with OpenCV VideoCapture to support real-time inference, visualization, and GPIO-based servo motor tracking for drone applications.

Face Detection System | Data Preparation, Data Preprocessing, YOLO-based Object Detection

- Built a real-time object detection system using YOLO on Raspberry Pi, capable of detecting multiple objects from live camera feed with optimized performance for edge devices.
- Integrated live video stream processing using Pi Camera.
- Achieved real-time detection under hardware constraints using YOLO model.

PROFESSIONAL EXPERIENCE

Indium Software **Mar 2025 - Aug 2026**

Associate Test Engineer *Bangalore*

- Supported game testing and client deliverables as a full-time Associate Test Engineer at iXie by Indium.
- Captured and validated gameplay footage for upcoming game events according to client requirements.
- Managed defect tracking and client requests in Jira while collaborating with QA teams and project managers.

EDUCATION

Jain College of Engineering **2020 - 2024**
Bachelor, Electronics and Communications Engineering *Belagavi*

COURSES & CIRTIFICATIONS

***Supervised Machine Learning: Regression and Classification** - Coursera (DeepLearning.AI & Stanford University)
***Advanced Learning Algorithms** - Coursera (DeepLearning.AI & Stanford University)
***Unsupervised Learning, Recommenders, Reinforcement Learning** - Coursera (DeepLearning.AI & Stanford University)
* **Data Science** Course by CodeWithHerry
* **Python Full-Stack Development** by PySpiders

EXTRA-CURRICULAR ACTIVITIES

2017

- Joined the elite group of Black Belts in Karate in 2017.
- State Championship: Silver Medallist in Departmental Sports Karate.
- National Championship: 2nd in National Karate Championships.
- Double victory! Won 1st place in both the Robo Race competition organized by JCER Belagavi and the Bot Building competition organized by Basaveshwar Engineering College, Bagalkot.

IEEE Women in Engineering (WIE) **2022 - 2023**
Chair

- In 2022-2023, I served as Chair of IEEE Women in Engineering (WIE). Also held the position of IEEE Media Head in 20212022.

Google Developer Groups, Belagavi (GDG) and Karnataka State Council For Science And Technology (KSCST) **2022**
Volunteered

- Volunteered at Google Developer Groups, Belagavi (GDG) in 2022 and Karnataka State Council For Science And Technology (KSCST), organized by VTU Belagavi.

LANGUAGES

- Kannada
- English
- Hindi
- Marathi